Exam practice 3

This section contains both past paper questions and practice questions covering a range of command words found in Paper 4: Geographical Investigations. You will find past paper questions or practice questions following the route to geographical enquiry, with example student responses and commentary. You will then answer similar practice questions or past paper questions. The questions draw together your knowledge and understanding of the route to geographical enquiry and will help you prepare for your assessment.

Understanding the route to geographical enquiry and planning fieldwork

When completing fieldwork, there are several steps you need to take to ensure that your investigation is reliable and unbiased. These steps must be followed in order and are part of the route to geographical enquiry, otherwise known as the scientific method.

Geographical enquiries begin with the formulation of hypotheses – statements that you will set out to accept, partly accept or reject. The following question tests your knowledge about forming hypotheses. It has an example student response and commentary provided. Work through the question first, then compare your answer to the example student response and commentary.

1 A group of students in Thailand studied how the characteristics of a river change downstream. The characteristics are shown in Figure 1.

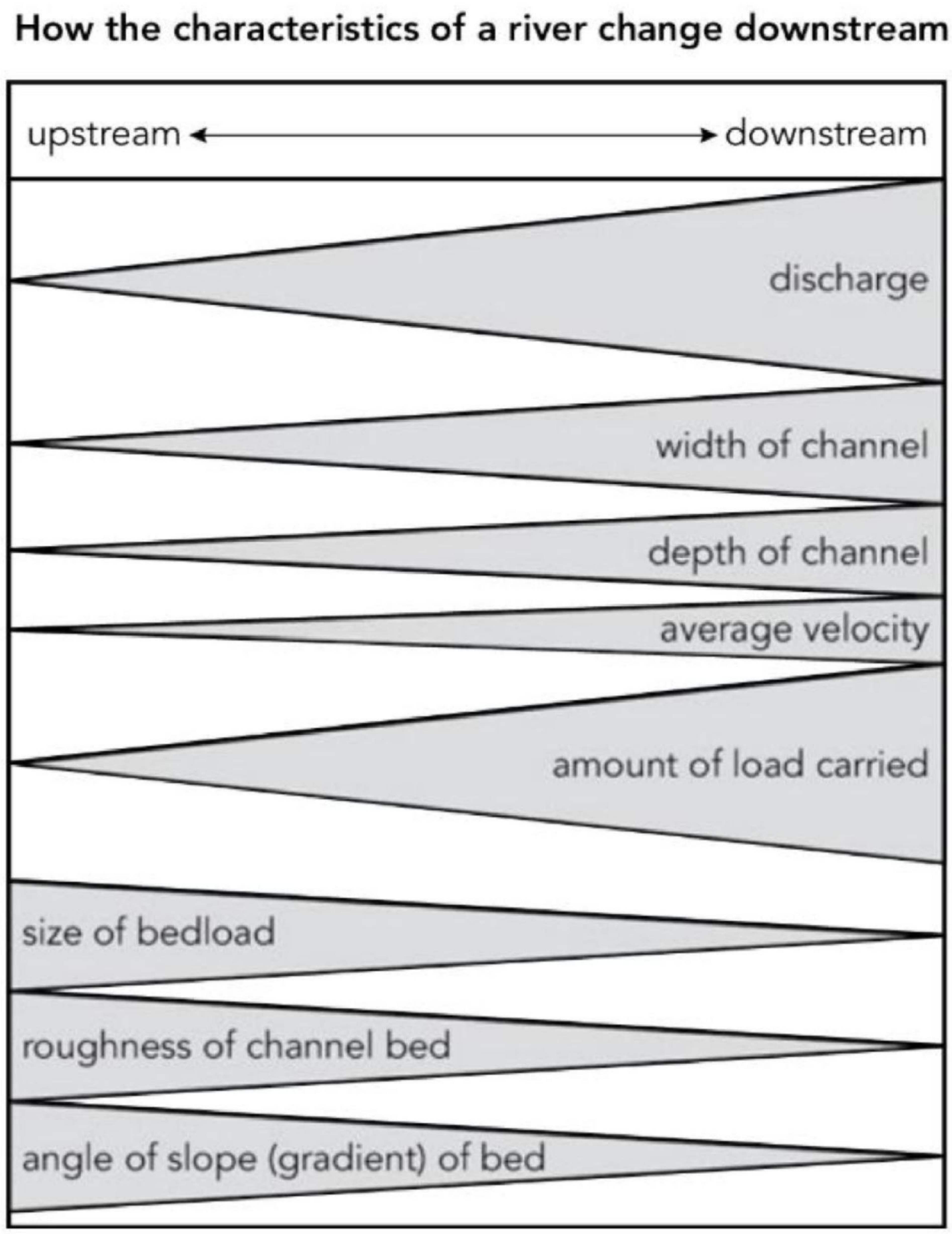

Figure 1

▶ **Image Description**

The students decided to investigate two characteristics by testing the following hypotheses at five sites along the River Huai Kup Kap:

Hypothesis 1: Pebbles on the riverbed (bedload) become smaller further downstream.

Hypothesis 2: The angle of slope (gradient) of the riverbed becomes less steep further downstream.

a Suggest a suitable hypothesis to investigate **one** other characteristic shown in the diagram. Do **not** choose size of bedload or angle of slope (gradient) of the riverbed. [1]

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Sample student response	Commentary
How wide the river is in different places.	This answer is incorrect as it is not phrased as a hypothesis. A hypothesis needs to be a testable statement, which makes a clear prediction about how a variable changes, for example: 'The width of the river increases further downstream.'
The size of bedload decreases downstream.	The question asked for a hypothesis that was not about the size of bedload or angle of slope. This answer is incorrect as the suggested hypothesis is about the size of bedload.

Sample student response	Commentary
The channel width increases with distance downstream.	This is a suitable hypothesis as it accurately selects and uses information from Figure 1 and is phrased as a testable statement.

Now try some more questions which test your knowledge about planning fieldwork and understanding the route to geographical enquiry, including Part b of Question 1 above.

b Describe a method to investigate your hypothesis at the five fieldwork sites. [4]

Cambridge IGCSE Geography (0460) Paper 41, Q2e(ii), June 2023

c Before they began their work, the students did a pilot study at a site on the river.

Identify **two** advantages of doing a pilot study from Table 1. Tick (✓) **two** choices. [2]

	Tick (✓)
draw a field sketch of the river's course	
get to know other students before they begin fieldwork	
learn how to work safely in the river	
practise fieldwork techniques	
look at different features along the river	

Table 1

Cambridge IGCSE Geography (0460) Paper 41, Q2a, June 2023

d To investigate **Hypothesis 1:** Pebbles on the riverbed (bedload) become smaller further downstream, the students selected 20 pebbles from the bed of the river at each site.

Which piece of fieldwork equipment from Table 2 did the students use when they measured the length of each pebble? Tick (✓) your choice. [1]

Equipment	Tick (✓)
callipers	
hygrometer	
quadrat	
ranging pole	

Table 2

Cambridge IGCSE Geography (0460) Paper 41, Q2b(i), June 2023

Data collection methods, fieldwork safety and equipment

Geographical enquiries require you to collect data. It is important to choose the right type of data so that it matches the aims of the investigation. Planning data collection methods requires an understanding of fieldwork safety and field equipment.

2 Students carried out fieldwork in their local town centre. They wanted to find out how successful the regeneration of the town centre had been.

The students decided to test the following hypotheses:

Hypothesis 1: Older people are happier with the regeneration than younger people.

Hypothesis 2: The regeneration has improved green spaces within the town centre.

Devise three suitable questions that the students could use in a questionnaire.

Sample student response	Commentary
1 Where do you live? 2 What is your name? 3 Do you like green spaces?	The first two questions are not needed to answer these hypotheses; in fact, questionnaires shouldn't ask for personal details like names at all. Question 3 is about green spaces, but it won't reveal whether the regeneration has improved green spaces within the town.
1 How old are you? 2 Do you like the regeneration that's taken place? 3 Do you visit green spaces?	The first question is very good. The second question is related to the hypotheses, but asking whether people 'like' the regeneration could get a different response than asking

Sample student response	Commentary
	people if they are ‘happy’ with the regeneration, which is what Hypothesis 1 is about. Question 3 is irrelevant and does not help to answer either hypothesis.
1 How old are you? 2 Are you happy about the regeneration that’s taken place here? 3 Do you think the regeneration has improved green spaces within the town?	Questions 1 and 2 are focused and clear. They will help to answer Hypothesis 1. Question 3 is also clear and will help to answer Hypothesis 2.

Now try some more questions which test your knowledge about data collection methods, safety and field equipment.

3 **a** Students carried out some fieldwork about tourism at two sites near Kuala Lumpur, the capital city of Malaysia. The sites at Kuala Lumpur Bird Park and Batu Caves are shown in Figure 2.

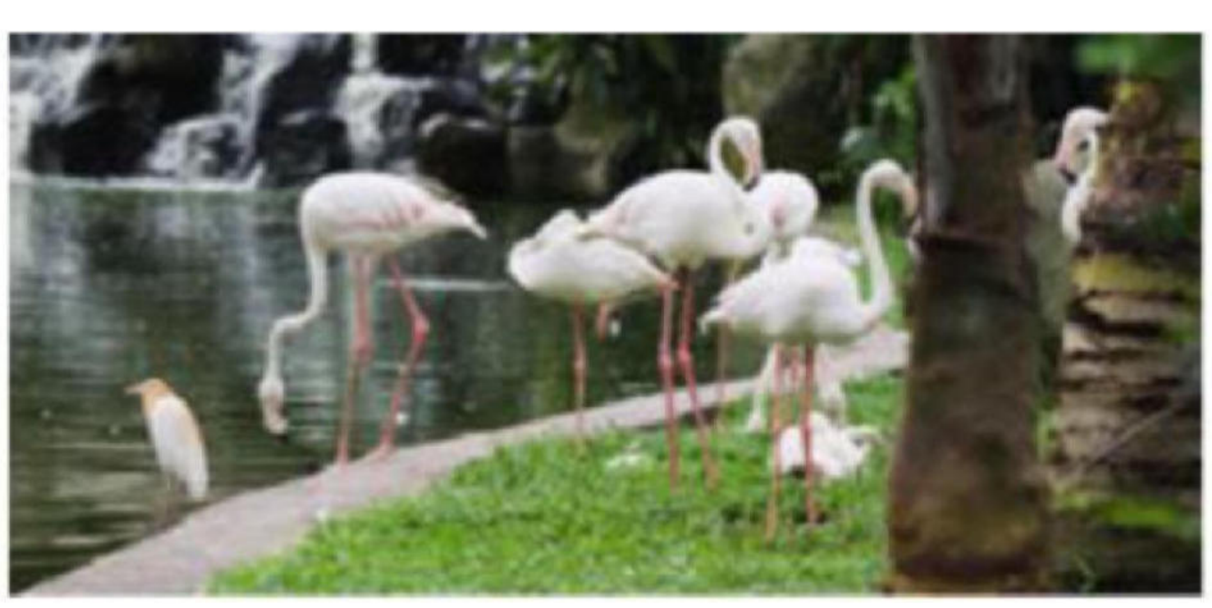

Figure 2

The students decided to test the following hypotheses:

Hypothesis 1: More foreign tourists come from Southeast Asia than from other parts of the world.

Hypothesis 2: Visitors to the Kuala Lumpur Bird Park spoil the environment more than visitors to the Batu Caves.

To test **Hypothesis 1** the students asked 100 visitors at the tourist sites which country they came from.

The students used a systematic sampling method to select people to question. Describe this method of sampling. [2]

b Why is sampling a useful fieldwork technique? [2]

Cambridge IGCSE Geography (0460) Paper 41, Q2a(i), a(ii), June 2024

4 Students did fieldwork on a local coastline. They investigated a variety of topics including longshore drift and coastal management.

The students agreed to test the following hypotheses:

Hypothesis 1: Longshore drift is occurring along the local coastline.

Hypothesis 2: Coastal defences have a positive impact on the local coastline.

To investigate Hypothesis 1: Longshore drift is occurring along the local coastline, some students used the fieldwork method described in Figure 3.

A fieldwork method to test Hypothesis 1

- Use brightly coloured waterproof paint to paint 50 pebbles of different shapes and sizes.

- Select a clear section of beach avoiding obstacles and groynes.
- Put the pebbles in the zone of swash and backwash and mark the position of the pebbles with a ranging pole.
- After one hour find the pebbles and work out the direction and distance that they have moved from the ranging pole.
- Collect the pebbles to use them again. Replace any missing pebbles with spare ones.
- Repeat the method three times.

Figure 3

Suggest why the students:

- painted the pebbles
- repeated their method three times. [2]

Cambridge IGCSE Geography (0460) Paper 42, Q1b(i), March 2024

5 When conducting fieldwork in an urban area, students identified several potential risks shown in Table 3. Suggest one precaution that the students could take to reduce each possible risk. [2]

Possible risk	Possible precaution
getting lost	
being run over by vehicles	

Possible risk	Possible precaution

Table 3

6 How are ranging poles used in geography fieldwork? [3]

7 **a** Students were planning fieldwork on a local beach. This is shown in Figure 4. The beach has groynes which go from the back of the beach towards the sea.

Figure 4

Before the students began their fieldwork, their teacher reminded them of the safety precautions they must take.

Table 4 shows four possible dangerous situations. Suggest one different precaution that the students could take to reduce **each** possible danger. [4]

Possible danger	Possible precaution
Heavy rain is forecast on the day of the fieldwork.	
High cliffs at the back of the beach.	

Possible danger	Possible precaution
Powerful waves break onto the beach.	
The beach is covered by the sea at high tide.	

Table 4

134

Cambridge IGCSE Geography (0460) Paper 42, Q1a November 2023

In Paper 4, there are often short questions that check how well you understand the topic. Knowing your topic well also means you can write sensible hypotheses and understand the aims of your fieldwork more clearly. It is therefore important to know your fieldwork topic well when doing your planning.

b Describe the beach shown in Figure 4. [2]

Cambridge IGCSE Geography (0460) Paper 42, Q1b, November 2023

Presenting and analysing data, and making conclusions

There are many ways to present data, and it is important to choose the right presentation technique. Analysing data involves looking for patterns and trends, and conclusions are reasoned judgements that use data to support them. Making conclusions also requires revisiting the hypotheses that you set out at the start of the investigation. The questions in this section test your understanding of presenting and analysing data and making conclusions.

8 Students have collected data on different types of land use (e.g. residential, commercial, industrial, recreational) along a transect from the centre of a town to its outskirts.

Suggest a method to present this information clearly and effectively. [4]

Sample student response	Commentary
I would use a bar chart to show the land use types.	This answer is too simplistic. Bar charts are a good idea, but it isn't clear whether the candidate is intending to draw more than one bar chart. As this data is taken along a transect, more than one bar chart needs to be drawn.
I would do a pie chart, showing the different land uses across the transect. To do this, I would add up all the land uses at all the sites and turn them into percentages. My pie chart would then summarise these percentages.	This answer has more detail, and pie charts are useful when showing percentages. However, this student intends to add up the land uses at all sites – this only tells us the overall land use in the city, and not how the land use changes across the transect.
I would draw a land use transect diagram, with the horizontal axis showing distance from the town	This answer is good as it uses an appropriate method of presentation with each site

Sample student response	Commentary
centre and coloured bands representing different land uses at specific sites along the route. This allows clear visual comparison of how land use zones change with distance, making it easy to identify patterns such as the transition from commercial to residential areas.	displayed separately. There is a strong explanation of why the method is effective for showing change and comparison. Finally, the use of colours will help with clarity.

Now try some more questions which test your knowledge about presenting and analysing data and making conclusions.

9 **a** Students did fieldwork on a local coastline. They investigated a variety of topics including longshore drift and coastal management.

The students agreed to test the following hypotheses:

Hypothesis 1: Longshore drift is occurring along the local coastline.

Hypothesis 2: Coastal defences have a positive impact on the local coastline.

The results of the students' measurements are shown in Table 5.

On a copy of Figure 5, **plot the results** of the total number of pebbles found and the average length of pebble that moved between 40.1 m and 50 m from the ranging pole (see PPQ 9a in the Resource Sheets). [2]

Cambridge IGCSE Geography (0460) Paper 42, Q1b(ii) March 2024

Distance moved east from ranging pole (m)	Total number of pebbles found in the three tests	Average length of long axis of pebbles (cm)
0.1 to 10	4	8.5
10.1 to 20	15	7.2
20.1 to 30	24	6.6
30.1 to 40	33	5.7
40.1 to 50	**21**	**5.5**
50.1 to 60	17	5.1
60.1 to 70	13	4.3
70.1 to 80	9	4.0
more than 80	3	3.4

Table 5

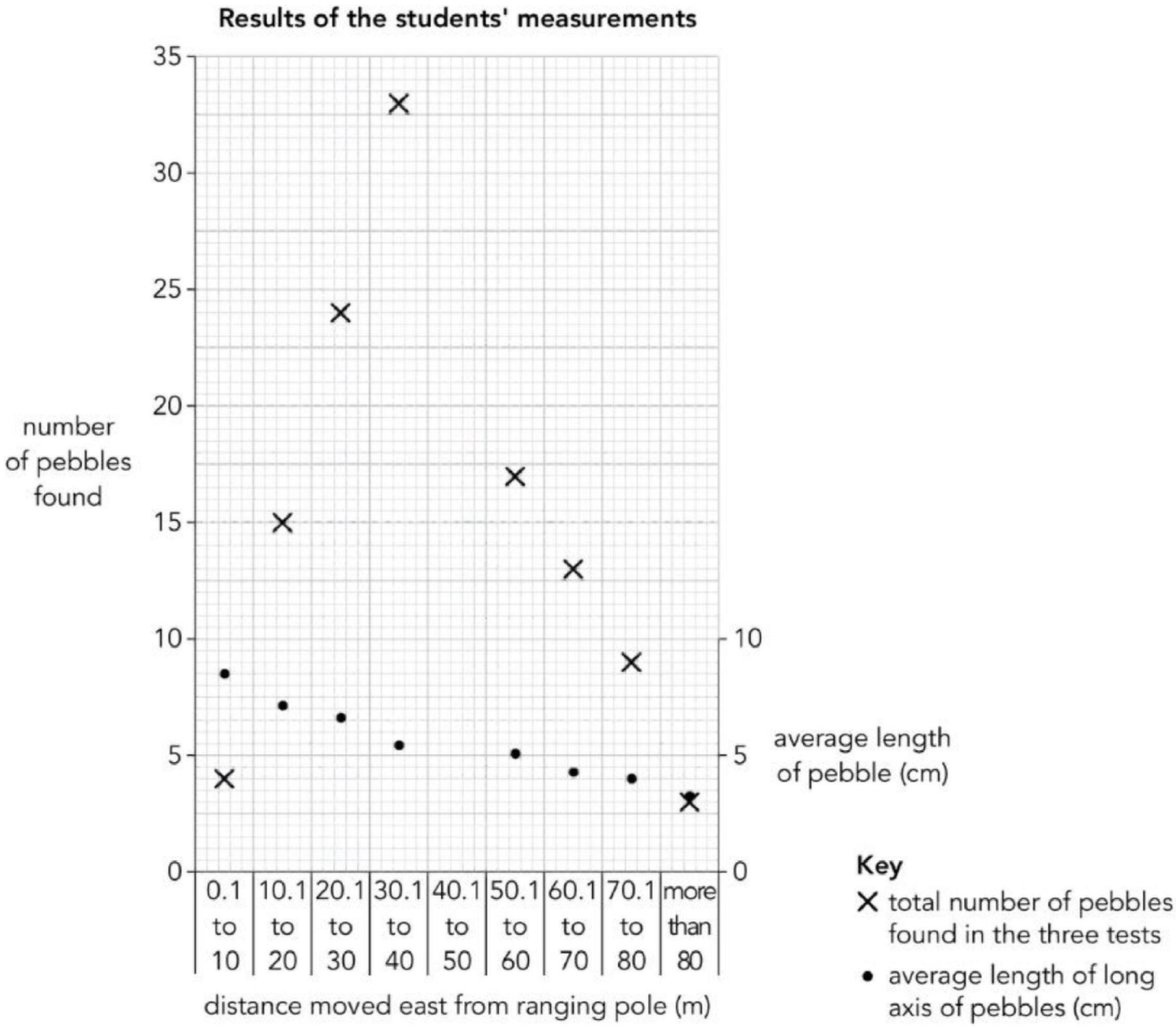

Figure 5

▶ **Image Description**

b Do the results shown in Figure 5 and Table 5 support Hypothesis 1: Longshore drift is occurring along the local coastline? Use data to support your conclusion. [4]

Cambridge IGCSE Geography (0460) Paper 42, Q1b(iii), March 2024

10 a Students did fieldwork on Ubin island, a small island between Malaysia and Singapore. Ubin is a rural area with little economic development. There has been much local discussion about whether the island should be protected from future development.

The students did their fieldwork at 20 sites on the eastern side of the island near to the main village.

They decided to investigate the following hypotheses:

Hypothesis 1: Environmental quality increases away from the village.

Hypothesis 2: Economic development on the island would bring more benefits than problems for local people.

To investigate **Hypothesis 1** one student completed a bi-polar survey on environmental quality at the different sites. The survey sheet is shown in Figure 6.

Site number:	**Score**					
Positive description	**+2**	**+1**	**0**	**–1**	**–2**	**Negative description**
beautiful landscape						ugly landscape
unspoilt by human activity, e.g. no litter						human activity spoils the landscape, e.g. litter
varied types of scenery						no variety of scenery
safe and appealing						unsafe and hostile

Site number:	**Score**					
Positive description	**+2**	**+1**	**0**	**–1**	**–2**	**Negative description**
peaceful						noisy
human development fits in with the natural environment						development by people does not fit in with the natural environment

Total environmental quality score:

Figure 6

Suggest how the students could make sure that their bi-polar survey results are reliable.

Adapted from Cambridge IGCSE Geography (0460) Paper 42, Q2b(ii), June 2022

b Table 6 shows Student A's results of the bi-polar survey for the 20 fieldwork sites.

Site number	**Attractiveness of landscape**	**Result of human activity**	**Variation of scenery**	**Safety and appeal**
1	0	–1	0	+1
2	–2	–1	–1	0

Site number	Attractiveness of landscape	Result of human activity	Variation of scenery	Safety and appeal
3	−1	−1	0	0
4	+1	−1	0	+1
5	0	−1	0	0
6	+1	0	+2	+1
7	0	0	0	+1
8	+2	+1	+2	+1
9	0	0	0	+1
10	0	0	−1	0
11	0	0	−1	−1
12	+1	0	+1	+2
13	0	−1	−1	0
14	−2	0	−2	−1
15	−1	−1	−1	−1
16	+1	0	+1	+1
17	0	0	−1	−1
18	+1	−2	+1	+1
19	+2	+1	+1	+2
20	+2	−2	+1	−1

Site number	Attractiveness of landscape	Result of human activity	Variation of scenery	Safety and appeal
Total score across all sites	+5	–9	+1	+7

Table 6

Which site has the lowest total environmental quality score? [1]

Cambridge IGCSE Geography (0460) Paper 42, Q2c(i) June 2022

c On a copy of Figure 7, plot the total environmental quality score for Sites 15 and 16 (see PPQ 10c in the Resource Sheets). [2]

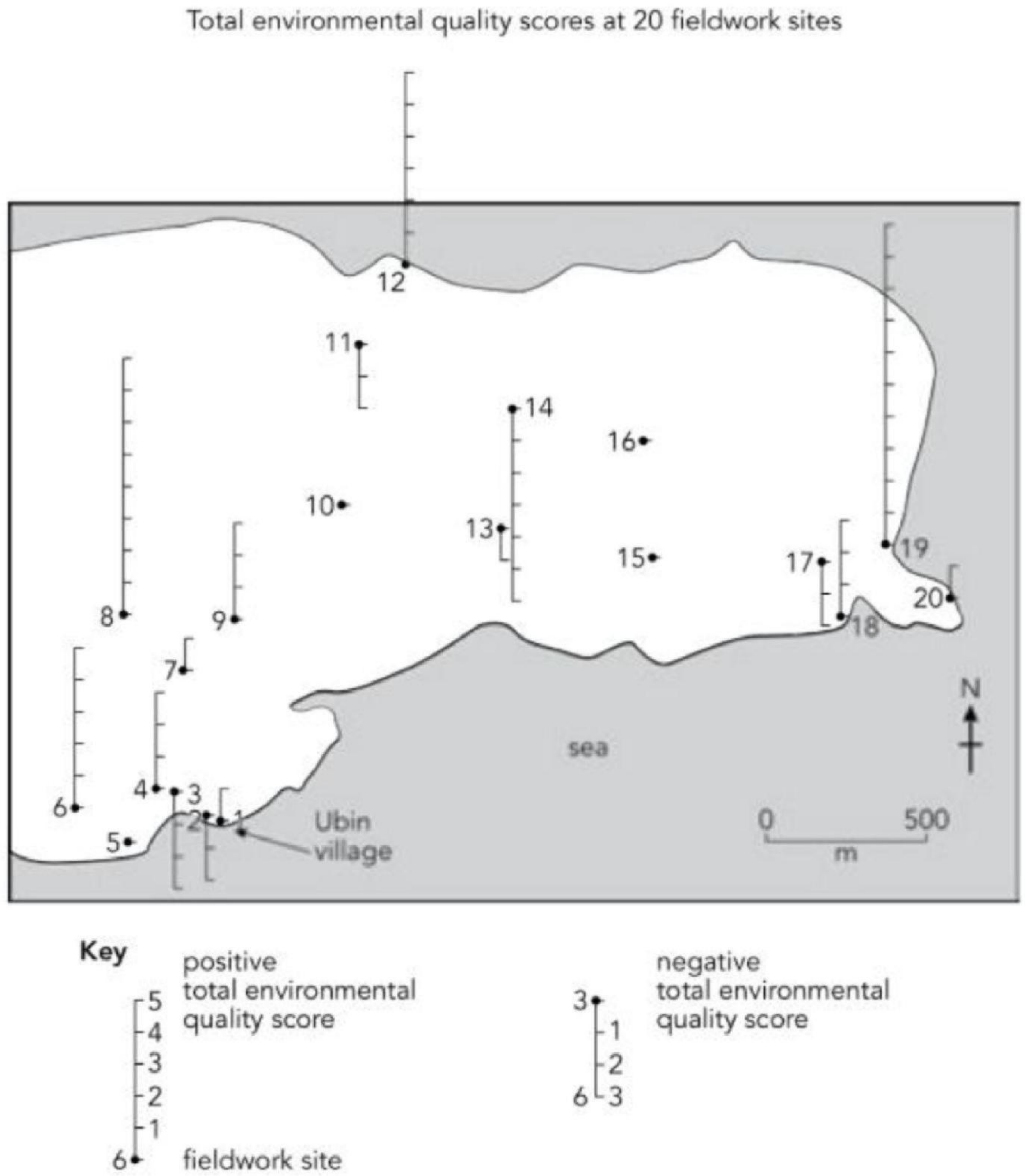

Figure 7

Adapted from Cambridge IGCSE Geography (0460) Paper 42, Q2c(iii), June 2022

d Estimate the distance between the two sites with the highest total environmental quality score (Sites 8 and 19).

Choose the best estimate from the options below:

i 500 km

ii 1000 km

iii 1500 km

iv 2200 km [1]

This part of the question is author-written

11 a A group of students visited Jwaneng, a large open-pit diamond mine in Botswana, an MIC in Africa. Most mining is done by blasting at or near the surface.

The students tested the following hypotheses:

Hypothesis 1: Employment is the most important benefit of the mine for residents of Jwaneng.

Hypothesis 2: The level of pollution increases towards the mine.

To investigate **Hypothesis 1** the students used a questionnaire with 100 local residents to study the impacts of Jwaneng mine. This questionnaire is shown in Figure 8.

Resident questionnaire

We are doing a survey about the local mine as part of our Geography fieldwork. Please will you answer the following questions?

1. What do you think are the benefits of Jwaneng mine?

....

....

....

....

2. What do you think are the disadvantages of Jwaneng mine?

....

....

....

....

Thank you for your time.

Figure 8

The results of Question 1 (What do you think are the benefits of Jwaneng mine?) are shown in Table 7. Use this data to **complete Figure 9** (see PPQ 11a in the Resource Sheets). [2]

Benefits of the mine	Number of answers
employment	76
medical facilities	44
shops	30
education facilities	32
recreation facilities	20
aeroplane runway and roads	15

Table 7

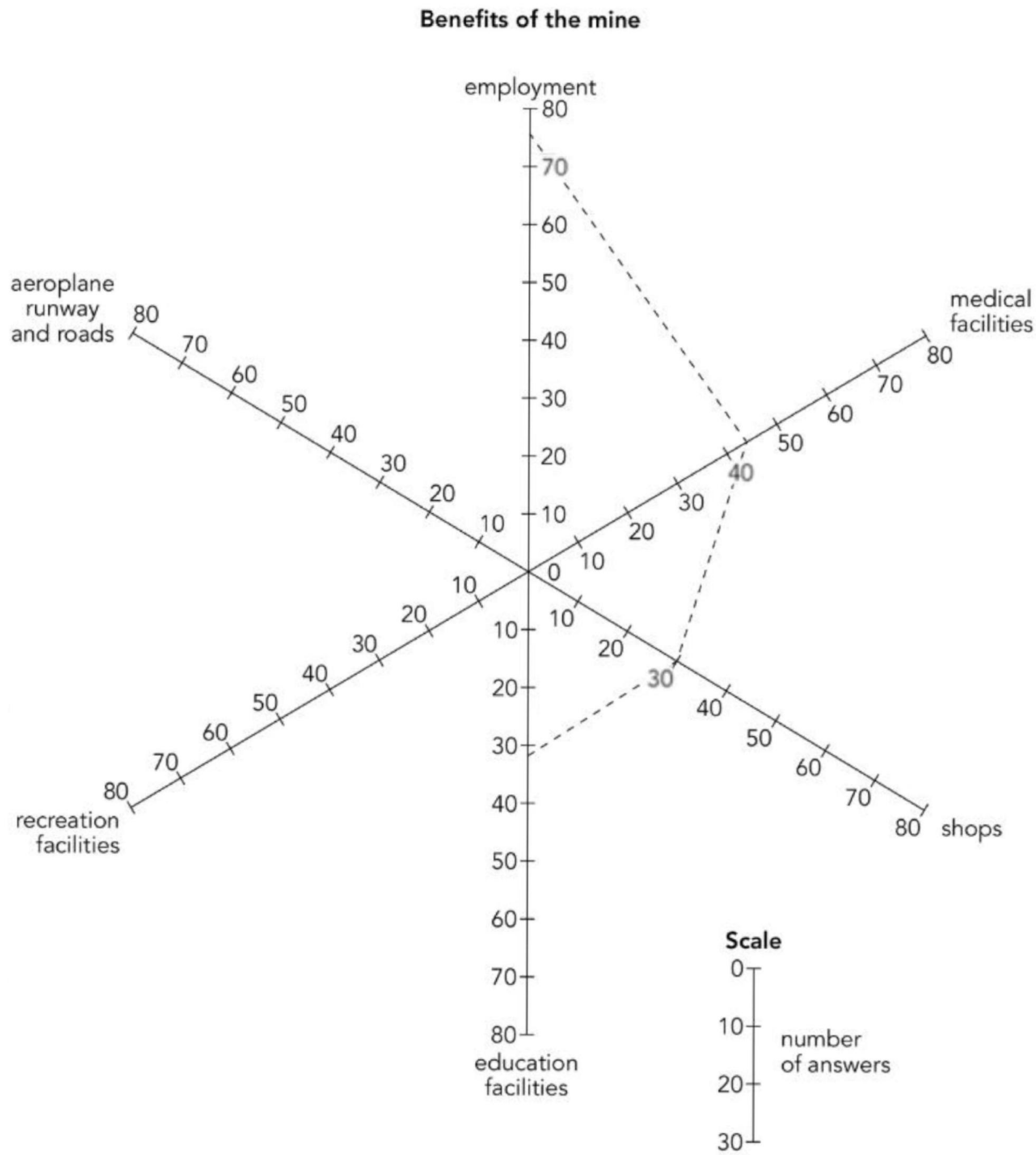

Figure 9

▶ **Image Description**

Adapted from Cambridge IGCSE Geography (0460) Paper 41, Q1c(iii), November 2022

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